University of Ottawa

Department of Mathematics and Statistics

MAT 1302D: Mathematical Methods II

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Second Midterm Exam – Solutions

March 18, 2010

Version A

Surname _____ First Name _____

Student # _____ DGD (1–4) _____

Instructions:

(a) You have 80 minutes to complete this exam.
(b) The number of points available for each question is indicated in square brackets.
(c) Unless otherwise indicated, you must justify your answers to receive full marks.
(d) All work to be considered for grading should be written in the space provided. The reverse side of pages is for scrap work. If you find that you need extra space in order to answer a particular question, you should continue on the reverse side of the page and indicate this **clearly**. Otherwise, the work written on the reverse side of pages will **not** be considered for marks.
(e) Write your student number at the top of **each page** in the space provided.
(f) No notes, books, scrap paper, calculators or other electronic devices are allowed.
(g) You are strongly advised to write in **pen**, not pencil.
(h) You may use the last page of the exam as scrap paper.

Good luck!

Please do not write in the table below.

Question	1	2	3	4	5	6	7	Total
Maximum	5	5	3	5	3	4	5	30
Grade								

1. **[5 points]** Calculate the determinant of the matrix $A = \begin{bmatrix} 1 & 0 & 4 & -2 \\ 2 & 5 & -3 & 0 \\ 3 & 0 & -1 & 2 \\ 1 & 1 & 1 & -1 \end{bmatrix}$.

2. **[5 points]** Is the matrix $B = \begin{bmatrix} 1 & -1 & 2 & 1 \\ 1 & 0 & 2 & 1 \\ -1 & 1 & 1 & -2 \\ 1 & -1 & 2 & 2 \end{bmatrix}$ invertible? If so, then find B^{-1}.

3. (a) **[1 point]** Let A and B be 4×4 matrices. You don't know what A and B are but you know that $\det A = -2$ and $\det B = 3$. Calculate $\det(A^T B^{-2} A B^T)$.

(b) **[2 points]** Let C and D be 5×5 matrices. You don't know what C and D are, but you know that

$$C^T D C^{-1} = \begin{bmatrix} -2 & 0 & 0 & 0 & 0 \\ 3 & -1 & 0 & 0 & 0 \\ 1 & 0 & 4 & 0 & 0 \\ 2 & -5 & 3 & -1 & 0 \\ 1 & 3 & 0 & -4 & -2 \end{bmatrix}.$$

Calculate $\det D$.

4. Let $A = \begin{bmatrix} 1 & -2 & 0 & -1 & 3 & -1 \\ 2 & -4 & 1 & 0 & 5 & 3 \\ -1 & 2 & 1 & 2 & -2 & 4 \end{bmatrix}$.

(a) **[3 points]** Find a basis for $\operatorname{Col} A$.

(b) **[1 point]** Determine the rank of A.

(c) **[1 point]** Determine the nullity of A. (Recall that the nullity of A is $\dim \operatorname{Nul} A$.)

5. Determine which of the following sets are subspaces of $\mathbb{R}^n$ for the given n. In each case, you should justify your answer.

(a) **[1 point]** $H = \left\{ \begin{bmatrix} x \\ -2x \\ x \end{bmatrix} \;\middle|\; x \geqslant 0 \right\}$, for $n = 3$.

(b) **[1 point]** $H = \left\{ \begin{bmatrix} x - y \\ 2x - y \\ -2y \end{bmatrix} \;\middle|\; x, y \text{ are arbitrary scalars} \right\}$, for $n = 3$.

(c) **[1 point]** $H = \left\{ \begin{bmatrix} x - 2y + 3 \\ -x + 2y + 3z \\ -1 \\ -z + y \end{bmatrix} \;\middle|\; x, y, z \text{ are arbitrary scalars} \right\}$, for $n = 4$.

6. An economy has two sectors: Manufacturing and Service. In order to produce one unit of output, Manufacturing requires .4 units from its own sector and .5 units from Service. On the other hand, Service requires .2 units from its own sector and .6 units from Manufacturing to produce one unit of output.

(a) **[1 point]** Write down the consumption matrix C for this economy.

(b) **[1 point]** Determine the intermediate demands if Service decides to produce 40 units and Manufacturing decides to produce 50 units.

(c) **[2 points]** Determine the production levels required to meet a final demand of 27 units from Manufacturing and 90 units from Service.

7. Let $A = \begin{bmatrix} 1 & -2 & 2 & 0 & -5 \\ 1 & -2 & 2 & 2 & -7 \\ -2 & 4 & -4 & 0 & 10 \\ 1 & -2 & 2 & 1 & -6 \end{bmatrix}$

(a) **[4 points]** Find a basis for $\operatorname{Nul} A$.

(b) **[1 point]** Determine the rank of A.

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